

SP201 Student Activity Booklet

Group Work Lab 1: Introduction to Stata, Identifying and Transforming Variables

In this group exercise, we would like you to create a “work preferences” variable using the syntax we learned in Lab 1.

As mentioned in class, one reason given for the gender pay gap is that women prefer to work at lower paying jobs because it makes it easier for them to balance work and family obligations. Using the *Understanding Society Lab Dataset Documentation*, find the variable in the dataset that you believe best represents this idea.

Name of variable chosen	Why?

As you will see, there are several variables you could use to measure preferences, for the rest of this lab, we will use `d_scopfamb`. How to recode this variable? You have at least two options here. First, you could just keep the categories in the original variable and recode the rest into missing. Second, you could combine some categories together.

To decide what to do, researchers often think about the underlying concept they are trying to measure. As you remember from our lecture on measurement, this relates to the **validity** of our measure. So, do you think that people that “disagree” with the statement chosen will act differently than those who chose “strongly disagree.” If so, you should keep all of these categories.

Another way to decide how to measure a variable is to look at other quantitative research to see what other researchers have done. In general, to be able to compare your results to those of other researchers, you want to try to recreate the same variables they used in their analysis.

In this case, researchers often create **categorical** variables from attitudinal measures like the one we are using here. Next, I would like you to recode `d_scopfamb` into a new variable with three categories

- 1 Agrees,
- 2 Neither agree/disagree
- 3 Does not agree



Note: This variable has too many categories to be shown automatically by codebook, which by default only shows the first four categories with a positive value. You can change this by telling Stata to show more variable categories. In this case, let's ask for Stata to show up to 20 categories. To do so, type `codebook <variable name>, t(20)`

The table below details again the six steps we need to recode a variable. Fill it with the syntax that you will use to recode our preference variable. Call the variable **pref2**.

Step	Stata code
1. Create a new variable called pref2	Hint: <u>generate</u> <new variable> = <old variable>
2. Recode the new variable	Hint: recode <new variable> (old values =newvalue1) (old values=new value2) (old values=new value3)
3. Look at new variable	Hint: codebook <new variable>
4. Label new variable	Hint: <u>label</u> <u>variable</u> < variable> "name of variable"
5. Label variable categories	Hint: <u>label</u> <u>define</u> <variable label definition name> 1"label value for category 1" 2 "labelvalue for category 2" 3"label value for category 3" <u>label</u> <u>value</u> <variable name> <variable labeldefinition name>
6. Check to make surerecoding was done correctly	Hint: codebook <old variable>, t(number) codebook <new variable>, t(number)

Along with using codebook, you can also ask Stata to create a table comparing where cases are allocated in our old and new variables. To do so, type tabulate d_scopfamb pref2, missing. Here, we include the subcommand , missing because we want to see where the missing cases have gone. If you have done this correctly, you will get the output below.

family suffers if mother works full-time	Family/Work balance preference				Total
	Agree	Neither a	Disagree	.	
missing	0	0	0	19	19
inapplicable	0	0	0	4,265	4,265
proxy	0	0	0	3,940	3,940
refused	0	0	0	27	27
don't know	0	0	0	81	81
strongly agree	2,674	0	0	0	2,674
agree	8,706	0	0	0	8,706
neither agree/disagree	0	12,189	0	0	12,189
disagree	0	0	10,744	0	10,744
strongly disagree	0	0	4,512	0	4,512
Total	11,380	12,189	15,256	8,332	47,157

Group Work Lab 2: Univariate descriptive statistics for categorical variables

In this group exercise, we would like you to look at whether men and women have different occupational categories using both a tabular and a graphical form to summarise this information descriptively.

In the lab in fact, we have only calculated descriptive statistics for our overall sample. Now, working in groups, generate and fill in the information in the table below.

(Hint: use the `if` command to generate statistics for men and women separately).

Table 1: Relative Frequency of Occupations by Gender, 2012-2014

Occupation	Men	Women	Overall sample
Large employers & higher management			4.55
Higher professional			8.37
Lower management & professional			27.73
Intermediate			13.61
Small employers & own account			10.19
Lower supervisory & technical			7.42
Semi-routine			18.09
Routine			10.05
Total			100

Source: Understanding Society, Wave 4.

Based on the results in the table above, do men and women in the sample work in different types of occupations? How can you tell?

Group Work Lab 3: Univariate descriptive statistics for continuous variables

In this group exercise, we would like you to look at how big the gender pay gap appear in our sample. Working in groups, please fill out the table and answer the questions below.

Table 2: Gross monthly pay of sample, overall and by gender

	Total sample	Females	Males
Mean	1,904.73		
Median	1,590.20		
Standard deviation	1,510.59		
IQR	1,600.00		
N	17,023		

Source: Understanding Society, Wave 4

Do men or women make more per month in our sample? By how much?

If you had to choose to present the typical monthly wage of men and women in our sample, would you use the mean or the median? Why?

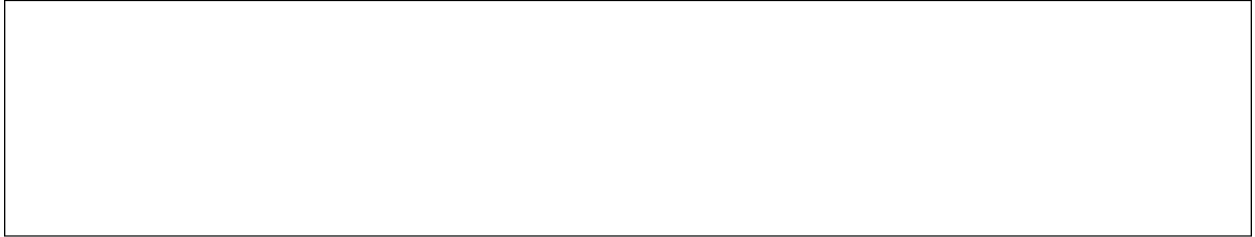
Now, you can also examine this in a graphical form as we learned in class. We can actually create graphs that detail separately, yet in the same graph, the graphical description for men and women in our sample

To generate separate histograms for men and women, type
`histogram pay, frequency by(female).`

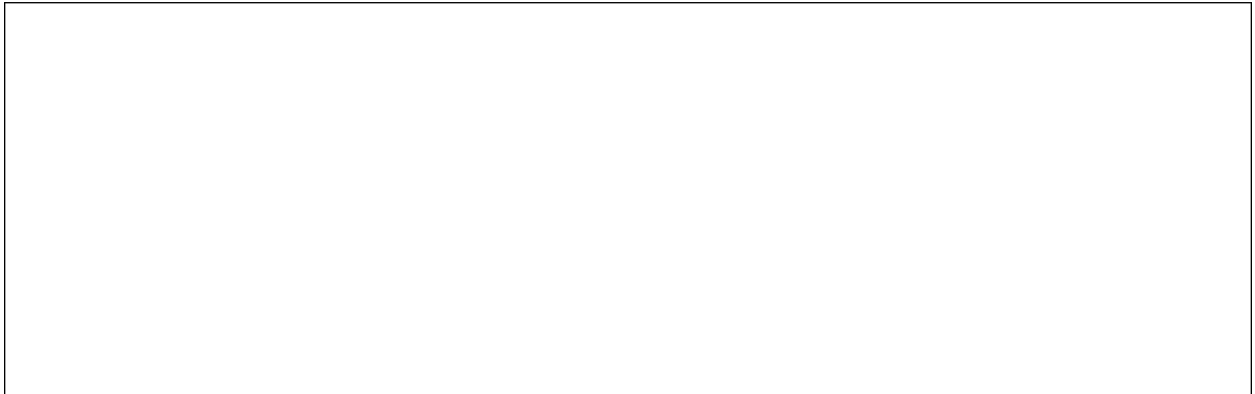
For a box plot, type
`box plot pay, by(female).`

Take a minute to create these graphs. Make sure to change the Titles to be informative and add information about the data. After you have done so, answer the following questions:

In this case, do you think it is clearer to present the gender pay gap as a table, histogram, or boxplot? Why?

A large, empty rectangular box with a thin black border, intended for the user to write their answer to the question about the best way to present the gender pay gap data.

Based on our results, do you think that gender pay disparities are a problem in the UK? Use information from the analysis you carried out above, as well as your knowledge of our dataset to justify your response.

A large, empty rectangular box with a thin black border, intended for the user to write their answer to the question about whether gender pay disparities are a problem in the UK, supported by their analysis and general knowledge.

Group Work Lab 4: Understanding inferential statistics

This week's group work consists in, as a group, reading and critiquing the following article. The aim is to see how academics and research write about their findings and present inferential statistics. To do so, we will review some of the sections of the following article.

Cacioppo, J., Cacioppo, S., Gonzaga, G., Ogburn, E., VanderWeelle, T. (2013). Marital satisfaction and break-ups differ across on-line and off-line meeting venues. *Proceedings of the National Academy of Sciences of the United States of America*, 110(25): 10135-10140.

Rather than reading the entire article, I'm just including the relevant sections here. After reading each section, please answer the associated questions.

First, read the abstract.

Abstract

Marital discord is costly to children, families, and communities. The advent of the Internet, social networking, and on-line dating has affected how people meet future spouses, but little is known about the prevalence or outcomes of these marriages or the demographics of those involved. We addressed these questions in a nationally representative sample of 19,131 respondents who married between 2005 and 2012. Results indicate that more than one-third of marriages in America now begin on-line. In addition, marriages that began on-line, when compared with those that began through traditional off-line venues, were slightly less likely to result in a marital break-up (separation or divorce) and were associated with slightly higher marital satisfaction among those respondents who remained married. Demographic differences were identified between respondents who met their spouse through on-line vs. traditional off-line venues, but the findings for marital break-up and marital satisfaction remained significant after statistically controlling for these differences. These data suggest that the Internet may be altering the dynamics and outcomes of marriage itself.

Now, answer the following questions:

- 1) **What is the research question?**

- 2) **What are the hypotheses?**

- 3) **What are the dependent and independent variables?**

- 4) **Do the findings seem relevant and important?**

Turning to the methods section, read the description on sampling below and answer the following questions.

Methods

The survey was conducted by Harris Interactive in June 2012. E-mail invitations to participate in an on-line survey were sent to 471,710 uSamp panelists. Of those who were e-mailed an invitation, 191,329 (40.06%) clicked on a Harris URL to electronically consent to take the survey. To determine eligibility, respondents were asked to specify: (i) whether they had married since 2005 (including 2005) (Yes/No), (ii) their year of birth, and (iii) their country of residence.

Eligibility criteria were that respondents resided in the United States, were at least 18 y of age, and reported being married at least once since the start of 2005. Of these 191,329 respondents, 122,265 were not eligible for the study based on these three criteria, 41,736 exited the survey early, and 7,207 were identified as fraudulent by uSample. Fraudulent responding was defined by Harris Interactive and its contractors as: (i) multiple surveys from the same respondent; (ii) surveys that are completed too quickly to reflect valid data from a human respondent; (iii) surveys from computers with IP addresses that were not within the United States; (iv) surveys sent from a geographical address that was not within the United States; (v) surveys originating from a known list of “professional survey takers”; (vi) a survey from a respondent who is registered multiple times within the uSamp panel; (vii) surveys completed by respondents using an open proxy, which allows users to conceal or disguise their IP address; and (viii) respondents who gave incorrect responses to a respondent instruction. If a respondent met any of these criteria, Harris Interactive categorized the respondent as “Not Qualified.” Of the remaining 20,121 initially classified as qualified, 74 were identified by Harris Interactive as invalid based on evidence of response biases, such as straight-line responding and inaccurate responding to catch- trials or inconsistent responding, leaving a final sample of 20,047 (15.28%) respondents.

Harris Interactive uses a weighting procedure based on propensity scores to be representative of the population of individuals married between 2005 and 2012. The demographics of the sample are summarised in Table S1. The analyses reported in the text are on weighted means and sample sizes. Harris Interactive sampled individuals, not couples, and slightly more men than women who served as respondents in this study reported meeting their spouses on-line. This finding suggests that the sample is an approximation rather than a perfect representation of the true population.

Now, answer the following questions:

- 1) What is the population of interest in this study? In other words, who do the authors claim the study results apply to?**
- 2) How were respondents recruited?**
- 3) Do you think that it is possible to say something about the population of interest based on this sampling strategy?**

Now, review the descriptive statistics presented in [Figure 1](#) and answer the following questions.

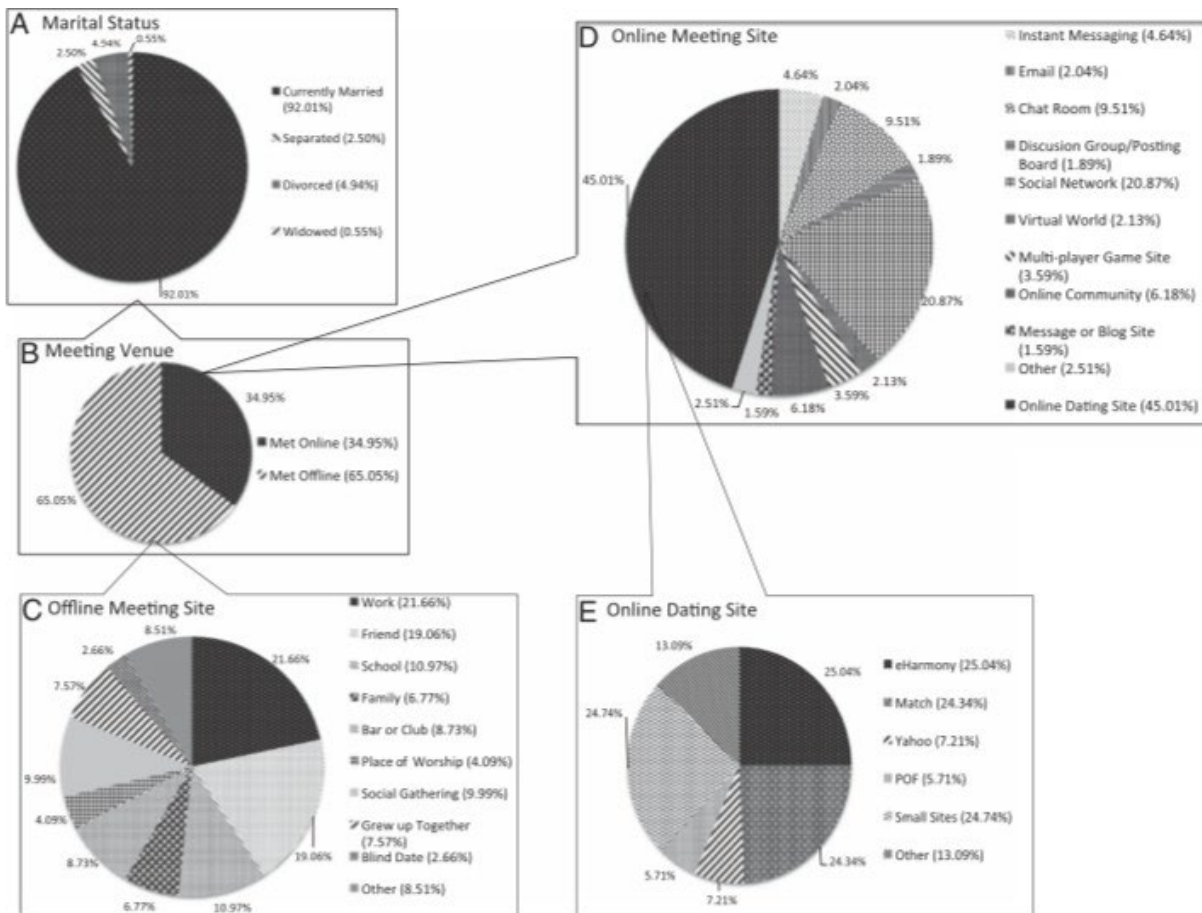


Fig. 1. (A) Marital status among the 19,131 (unweighted) respondents. (B) Meeting venue. (C) Offline meeting site. 21.66% of the respondents who met their spouse offline met through work, 19.06% through friends, 10.97% at school, 6.77% through family, 8.73% at a bar/club, 4.09% at a place of worship, 9.99% at a social gathering, 7.57% grew up together, 2.66% met on a blind date, and 8.51% met through "other" venues. (D) Online meeting site. Of the respondents who met their spouse online, 4.64% met through instant messaging, 2.04% through e-mail, 9.51% in a chat room, 1.89% through a discussion group/posting board, 20.87% through social network, 2.13% in a virtual world, 3.59% on a multiplayer game site, 6.18% in an online community, 1.59% on a message/blog site, 45.01% through an online dating site, and 2.51% met through "other" online venues. (E) Online dating site. Of the 45.01% who met through an online dating site, 25.04% met on eHarmony, 24.34% on Match, 7.21% on Yahoo, 5.71% on Plenty of Fish (POF), 24.74% were spread in smaller numbers (<100) across the remaining 14 dating sites specified in the survey (labeled hereafter as "small"), and 13.09% met on a dating site they specified as "other."

1) What is the purpose of this figure in the article?

2) Is Figure 1 the best way to present the data? If not, how could the presentation be improved?

Now, look at the descriptive statistics presented in [Table S1](#).

Table S1. Demographic characteristics for unweighted sample, weighted sample for marriages between 2005 and 2012, and the general United States married population across all years

Demographic	Sample characteristics		General United States married population estimates (%)
	Unweighted	Weighted	
<i>n</i>	19,131	19,038	
Percent female	62.96	52.70	50.00
Age (y)			
Mean	34.31 (10.69)	37.83 (11.77)	
18–29	39.97%	24.66%	8.62
30–39	34.96%	38.20%	19.08
40–49	14.17%	20.37%	22.55
50–64	9.46%	13.37%	31.82
65+	1.45%	3.39%	17.92
Ethnicity			
White/Caucasian	74.49%	67.83%	79.98
Black/African American	5.73%	8.56%	7.20
Hispanic	10.34%	18.29%	12.82
Asian or Pacific Islander	5.28%	2.77%	
Other	2.78%	1.39%	
Income			
Less than \$15,000	4.91%	3.68%	4.30
\$15,000 to \$24,999	8.58%	6.35%	6.60
\$25,000 to \$34,999	11.03%	7.88%	8.40
\$35,000 to \$49,999	16.55%	12.12%	12.60
\$50,000 to \$74,999	22.13%	19.68%	20.00
\$75,000 to \$99,999	15.13%	14.85%	15.80
More than \$99,999	16.77%	31.16%	32.30
Education			
High school or less	15.78%	23.66%	40.40
Associates or job training	15.35%	17.56%	9.70
College	50.05%	45.77%	37.20
Graduate school	18.82%	13.02%	12.70
Religious affiliation			
Catholic	19.70%	23.56%	
Christian/Protestant	39.81%	38.26%	
Jewish	2.24%	2.84%	
Mormon	2.81%	2.09%	
Spiritual, but unaffiliated	14.88%	14.84%	
Atheist/no religion	10.90%	9.84%	
Other religion	9.67%	8.57%	
Employment status			
Used full or part time	71.72%	75.04%	63.00
Retired	4.18%	6.59%	
Student	11.87%	8.53%	
Stay at home parent	28.81%	23.80%	

Harris Interactive applied weights to increase the representativeness of the survey data to the United States population of couples married (*Methods*). Estimates for the general United States married population were derived from the March 2011 US Census Current Population Survey. For Ethnicity in the 2011 US Census Current Population Survey, White/Caucasian included all races except Black/African American and Hispanic. Small percentages of participants did not report ethnicity (1.2%) or Income (4.3%) so column totals do not add to 100%. For Employment Status in the 2011 US Census Current Population Survey, estimates did not include reports of retired, student, or stay at home parents.

1) What is the purpose of this table?

2) Is this the best way to present the data? If not, how could the presentation be improved?

Table S6 presents study results about whether there is a relationship between the mode of meeting your spouse (on-line versus off-line) and differences in marital breakup. The main finding is presented in the last 3 rows of the table.

Table S6. Mean differences in rates of marital break-ups across different on-line and off-line meeting venues: Marriage as the unit of analysis

Meeting source	Weighted <i>n</i> (marriages)	% Marital break-up
On-line sources		
Instant messaging	343	8.53
E-mail	144	1.98
Chat room	664	4.23
Discussion group	138	8.69
Social network	1,544	5.42
Virtual world	156	10.77
Multiplayer game	257	5.09
On-line community	428	3.52
Message on blog	109	3.42
On-line dating	3,188	5.76
Other	190	4.53
χ^2 (<i>df</i>)		15.33 (10)
On-line dating sites		
eHarmony	791	3.66
Match	775	7.50
Plenty of Fish	201	9.75
Yahoo	227	5.48
Small sites	777	4.70
Other	417	6.70
χ^2 (<i>df</i>)		9.38 (5)
Off-line sources		
Work	2,858	7.04
Friends	2,503	8.49
School	1,410	5.61
Family	899	6.96
Bar/club	1,190	6.64
Place of worship	546	6.99
Social gathering	1,333	7.62
Grew up together	1,015	5.80
Blind date	363	6.42
Other	1,118	8.38
χ^2 (<i>df</i>)		10.07 (9)
On-line	7,051	5.87
Off-line	12,998	7.73
χ^2 (<i>df</i>)		10.48 (1)*

* $P < 0.01$. Named on-line dating sites are placed in alphabetical order.
 "Small sites" in the on-line dating category were named but did not produce more than 100 marriages.

Answer the following questions.

- 1) What percentage of relationships that start on-line break-up?
- 2) What percentage of relationships that start off-line break-up?

- 3) **Based on the p-value of the χ^2 test, is this difference statistically significant?**
- 4) **Does this difference in values seem substantively important? Why or why not?**

Lastly, read the Conclusion of the article:

Conclusion

In conclusion, marital outcomes are influenced by a variety of factors. Where one meets their spouse is only one contributory factor, and the effects of where one meets their spouse are understandably quite small and do not hold for everyone. The results of this study are nevertheless encouraging, given the paradigm shift in terms of how Americans are meeting their spouse. The present results addressed marital outcomes in the first 6 or 7 y of marriage, and longer-term follow-up studies are important to determine whether the observed differences in marital outcomes intensify or dissipate over even longer periods of time.

Although our analyses concern American marriages, the rapid increase in the use of the Internet is a global phenomenon. The mechanisms suggested above as contributing to our findings may not be specific to America, so investigations are needed to determine whether marriages that begin on-line, in contrast to off-line, predict better marital outcomes in other countries and traditional societies. What is clear from this research is that a surprising number of Americans now meet their spouse on-line, meeting a spouse on-line is on average associated with slightly higher marital satisfaction and lower rates of marital break-up than meeting a spouse through traditional (off-line) venues, and on-line venues are not as homogeneous as thought in terms of marital outcomes. Indeed, the present study shows that the tendency in past studies to treat all on-line venues as the same is no longer empirically justified.

Now, answer the following questions:

- 1) **What do the authors state are the main findings of their study in relation to their research questions? Please think about how the sampling strategy relates to their ability to address these questions.**
- 2) **Do these statements accurately reflect what is in the results section? Why or why not?**

3) What do the authors see as the relevance of their results? Do you agree? Why or why not?

Group Work Lab 5: Creating and interpreting contingency tables and the chi-square test

For this week's group work we would like you to create contingency tables for the relationship between gender and occupation and interpret the output.

In the lab, we have determined that women are more likely to be low-paid than men. So, now we can ask ourselves whether or not this can partly be attributed to the fact that men and women work in different occupations.

Before getting into Stata, code, take a moment to write your research question and your null and alternative hypotheses.

Research question	
H ₀ :	
H _a :	

Now, produce a contingency table that allows you to assess whether there are occupational differences between men and women in our sample. **Write the Stata code that you would use here.**

Looking over the table generated, are there certain occupations that seem more prevalent among women? What about men?

We want to test our hypotheses to determine whether there is a relationship between occupation and gender in the UK population. Write the Stata code to do so here:

Based on the results, do you reject your null hypothesis? Write the results of your analysis here in a way that conveys substantive, as well as statistical significance.

Now, are you meeting the assumptions of a chi-square test with this analysis? Write your response and the Stata code you would write to make sure you are doing so below.

Group Work Lab 7: Creating and interpreting scatterplots and correlation coefficients

This week's group work is intended to allow you to try to create your first transformed variable. We said in class that given the ubiquity of Pearson's correlation coefficient, researchers sometimes transform their variables to make sure they can meet the assumptions of the test rather than use Spearman's rho. The most common transformation used for a skewed income variable is to take its natural logarithm.¹

This is easy to do in Stata, simply type `generate log_pay=log(pay)`. Using the `histogram` command that we learned in lab 3, compare the distributions of the logged and unlogged versions of pay. **How has the distribution changed?**

Now, calculate Pearson's correlation coefficient using the new logged variable. Based on the output, what has changed? What would you say about the direction of the relationship between logged pay and hours? Is the relationship still statistically significant?

Now, calculate Spearman's rho using the new logged variable. Based on the output, what has changed? What would you say about the direction of the relationship between logged pay and hours? Is the relationship still statistically significant?

¹ If you need a refresher about what a logarithm is – or are just interested in what Stata is doing here, more details are provided at: <http://www.mclph.umn.edu/mathrefresh/logs.html>

Group Work Lab 8: Introduction to linear regression

For this week's group work your task is to assess whether the relationship between gender and wages still exists, after controlling for any differences in work hours, occupation, and preferences. At the end of this step, we will be able to say whether pay differences that exist between men and women can be accounted for by the fact that women work fewer hours than men, in different sorts of occupations, and different preferences.

Before we start using Stata here, let's just spend a minute thinking about what you are doing.

What is the research question and hypotheses in this analysis?

What is the regression equation for this model?

HINT: You do not need to include a regression coefficient (bx) for the reference group(s) in your analysis.

Now, write the Stata syntax needed to run this regression model.

HINT: remember to include `i.` in front of any categorical variables that you include.

Write a paragraph explaining your regression results. Remember, to interpret: the F- statistic, R-squared, and regression coefficient. Also, take the time to calculate the predicted wages of men and women who work 40 hours per week at large employers and higher management positions who disagree with the statement that families suffer when mothers work full-time.

In the regression that we ran above, the coefficients for our *Preferences* variable were quite different. Thinking about what you have learned, how could you explain these differences?

Group Work Lab 9: Introduction to Logistic regression

For this week's group work, you have to assess whether the relationship between gender and low pay still exists, after controlling for any differences in work hours, occupation, and preferences. At the end of this step, we will be able to say whether differences in low pay that exist between men and women can be accounted for by the fact that women work fewer hours than men, in different sorts of occupations, and hold different preferences.

What is the research question and hypotheses in this analysis?

What is the regression equation for this model?

HINT: You do not need to include a regression coefficient (b_x) for the reference group(s) in your analysis.

Now, write the Stata syntax needed to run this regression model. HINT: remember to include `i.` in front of any categorical variables that you include.

Based on the output obtained, is there still a significant difference in the likelihood that men and women will be low-paid. Although in the summative, you will focus your interpretation on the gender coefficient, take a moment here to write a sentence interpreting the coefficients on gender, work hours, occupation and preferences. For the occupation and preferences variables, you will need to pick the coefficient of one of the categories to interpret here.

Now, you will use the margins command to get predicted probabilities so we can think about the substantive findings. Please obtain the predicted probabilities for men and women who: work 40 hours per week, work in higher professional positions, and hold non-traditional gender roles. Write your interpretation of results below.

END